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Application of a Semi-Supervised One-Sided Adversarial Domain Generalization Framework in Face Anti-Spoofing

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Application of a Semi-Supervised One-Sided Adversarial Domain Generalization Framework in Face Anti-Spoofing

[Abstract**]** With the development of digital society, facial recognition systems face serious threats from presentation attacks (such as printed photos, video replays, 3D masks, and AI-generated faces). Enhancing the generalization capability of facial anti-spoofing (FAS) models under rapidly evolving and unknown attack scenarios has become a key research focus. Existing domain generalization methods (e. g., single-source domain generalization FAS models like SSDG) rely solely on labeled data from multiple source domains and fail to fully utilize unlabeled samples from the target domain. Although semi-supervised domain adaptation approaches can leverage information from the target domain, they are often limited to single-source domain settings.

To address the aforementioned issues, this paper proposes a semi-supervised one-sided adversarial domain generalization framework. Building upon SSDG, it incorporates a small number of labeled samples and a large number of unlabeled samples from the target domain. By jointly optimizing the unlabeled sample image block feature consistency detection module with labeled classification loss, unlabeled entropy minimization loss, domain adversarial loss, and real sample center alignment loss, the model's cross-domain generalization performance is further enhanced.

Experimental results on the source domains CASIA-FASD, MSU-MFSD, and Idiap Replay-Attack, as well as the target domain OULU-NPU, demonstrate that our method reduces the half total error rate (HTER) from 26% of the original SSDG to 9.2%, while improving the AUC to 96.89%. Ablation studies validate the effectiveness of each semi-supervised component and the image patch data augmentation module. Furthermore, the model exhibits strong generalization performance in detecting novel AI-generated fake faces. Our approach provides an effective solution for leveraging unlabeled data from the target domain to enhance the generalization capability of face liveness detection.

[Keywords**]** Face anti-spoofing; Domain generalization; Semi-supervised learning; One-sided adversarial; Entropy minimization

The Application of Semi-supervised Unilateral Adversarial Domain Generalization Framework in Face Anti-spoofing

Abstract: As digital society advances, face recognition systems face growing threats from presentation attacks (e.g., printed photos, video replay, 3D masks, AI-generated faces). Enhancing the generalization of face anti-spoofing (FAS) models under unseen attack scenarios is critical. Existing domain generalization methods like SSDG use only labeled multi-source data, ignoring unlabeled target samples. Conversely, semi-supervised domain adaptation methods leverage target information but mostly in single-source settings.

To address this, we propose a semi-supervised unilateral adversarial domain generalization framework built on SSDG. It introduces a few labeled target samples and many unlabeled ones, jointly optimizing a patch-shuffling consistency module, labeled target classification loss, entropy minimization, domain adversarial loss, and genuine-sample center alignment loss.

Experiments on CASIA-FASD, MSU-MFSD, Idiap Replay-Attack (source), and OULU-NPU (target) demonstrate that our method reduces HTER from 26% (SSDG) to 9.2%, boosts AUC to 96.89%, and generalizes well to AI-generated faces. Ablation studies validate the effectiveness of each component. This work effectively leverages unlabeled target data for FAS generalization.

Keywords: Face anti-counterfeiting; Domain generalization; Semi-supervised learning; Single-sided adversarial; Entropy minimization

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1 Introduction

1.1 Research Significance

In today's digital and information-driven society, traditional physical devices used for identity authentication are gradually being replaced by non-contact biometric authentication methods [1]. Facial recognition, due to its convenience and high accuracy, has achieved significant success both in theoretical research and practical applications, deeply integrating into everyday scenarios such as mobile device login, platform payment verification, and access control in public spaces. However, the widespread adoption of facial recognition has created new, easily accessible channels for information theft. This has led to a variety of spoofing attacks, including 3D mask printing, makeup and tattoo techniques, and printed replay attacks. Malicious attackers can easily launch various physical presentation attacks during the face capture phase and subsequently compromise facial recognition systems during the identification stage [2], posing serious security risks. As a result, the need to distinguish between real and fake faces—known as Face Anti-Spoofing (FAS) technology—is growing increasingly urgent.

To counter various spoofing attacks, numerous facial anti-spoofing methods have been proposed in the field, including traditional approaches based on texture and temporal cues for extracting authenticity indicators, deep learning-based masking and depth map techniques, as well as high-precision multimodal fusion algorithms. These methods have achieved significant discrimination performance against known attack types in conventional domains. However, with the rapid advancement of AI technologies in recent years, the emergence of AI-generated deepfakes—particularly those created using diffusion models and generative adversarial networks (GANs)—has led to smarter, more covert attack methods and a more severe crisis of portrait infringement, such as AI face-swapping, facial reenactment, and synthetic face generation. Traditional vulnerabilities exploited by existing methods—such as screen reflections, printed texture patterns, or unnatural 3D mask materials—may completely disappear in high-quality AI-generated content, rendering conventional liveness detection models largely ineffective. Meanwhile, large-scale generative models are evolving rapidly, further advancing facial forgery techniques.

As technology continues to advance, traditional deep learning methods for detecting AI-generated facial features against real ones may become ineffective in the face of newer and more sophisticated AI-based facial generation attacks.

To achieve high-precision anti-spoofing performance with limited and lightweight labels under novel attacks, it is essential to enhance the model's generalization capability and reduce its data dependency. This paper employs a domain generalization framework that enables the model to learn universal and intrinsic features of real human faces, thereby detecting "non-living" anomalies based on these shared characteristics. This allows the model to maintain strong discriminative ability even against unseen attack types and acquisition environments during training. By incorporating a semi-supervised learning mechanism, we further alleviate the heavy reliance of traditional domain generalization methods on large-scale, multi-source datasets with fine-grained domain labels, as well as the dependence of unsupervised learning on abundant target-domain data. This enables dual lightweight training across source and target domains, significantly reducing the cost and difficulty of annotating new attack data. The approach ensures strong generalization, high data efficiency, and strong interpretability, thereby enhancing the proactive defense capabilities and reliability of face recognition systems against emerging deepfake attacks involving AI-generated faces. The proposed semi-supervised domain generalization fusion method provides effective technical support for building more robust, practical, and low-cost facial anti-spoofing systems.

1.2 Current Research Status at Home and Abroad

1.2.1 Traditional methods based on handcrafted features

Early facial anti-spoofing techniques prior to 2018 were primarily based on traditional handcrafted feature methods [3], which relied heavily on extensive prior knowledge of human liveness signals and could be broadly categorized into approaches based on facial texture cues, motion, and temporal information. Methods based on facial texture cues mainly investigated differences in image variations between real and fake faces in color space by employing classical handcrafted descriptors (such as LBP, SIFT, SURF operators). Representative studies include de Freitas Pereira et al. [4], who introduced a novel LBP_TOP operator that integrated spatial and temporal information into a single framework.

Texture descriptors; motion-based anti-spoofing detection methods include analyzing blinking, facial and head movements, annotation tracking, and remote physiological signals (such as rPPG [5]).

but current live signals are now easily generated by AI

Technical breakthroughs may require extracting data from prolonged interactive facial videos, making rapid deployment of recognition difficult.

1.2.2 Fully supervised method based on deep learning

With the advancement of deep neural networks, face anti-spoofing methods based on deep neural networks have gradually replaced traditional hand-crafted feature approaches due to their strong feature representation capabilities, becoming the mainstream in the field. Researchers have integrated traditional features with deep neural networks, improving face spoofing detection performance through techniques such as cascaded fusion and feature-level fusion [6]. However, conventional deep supervised methods perform well on single datasets but suffer significant performance degradation when facing domain shifts across different scenarios and attack types. To address this issue, three main solutions have emerged in the field.

a. Multimodal FAS learning. Single RGB modality provides limited information and struggles to handle complex attacks. Multimodal approaches integrate data from multiple sensors such as RGB, depth, and infrared, leveraging the complementary nature of different modalities to enhance detection robustness. Since 2023, frameworks combining Vision Transformers (ViT) with multimodal data—such as MA-ViT [7]—have emerged to improve model generalization and adaptability. Daiyuan Li et al. [8] proposed a text-guided dropout token module that selects semantically invariant token blocks for multimodal fusion, further improving cross-validation accuracy. However, practical deployment of multimodal methods still faces challenges including high computational demands and significant distributional differences across modalities. In 2026, lightweight encrypted multimodal fusion biometric recognition schemes were introduced [9], yet their application in anti-counterfeiting detection remains limited and has not been widely adopted.

b. FAS based on spoofing trace disentanglement/decoupling learning. Decoupling learning aims to separate the general spoofing features from irrelevant attributes such as identity, expression, and illumination, which may vary across different datasets in facial images.

By separating the spoofing traces, the model's generalization capability against unknown attacks is enhanced. Jourabloo et al. [10] first proposed a Spoofing Trace Separation Network based on noise modeling, which decomposes forged facial inputs into spoofing noise and authentic faces, thereby improving model interpretability. Liu et al. [11] further extended this to a Spoofing Trace Disentanglement Network, capable of simultaneously reconstructing both genuine and synthetic forged faces, significantly enhancing robustness against various types of attacks.

c. Domain generalization methods. To further enhance generalization performance, an increasing number of researchers have introduced transfer learning ideas, leveraging domain generalization strategies to improve a model's cross-domain adaptability. Domain generalization aims to train models that can generalize directly to unseen domains without access to target domain data, using multiple source domains. These approaches can be broadly categorized into three subtypes: feature alignment, style alignment, and gradient alignment. From the perspective of feature alignment, Shao et al. [12] applied a dual-force triplet mining constraint to tightly cluster features of similar samples while sufficiently separating those of dissimilar ones, thereby designing a universal feature extractor adaptable across multiple source domains. The SSDG framework [13] adopted in this paper builds upon triplet-based methods by introducing a "one-sided adversarial" mechanism, aligning only the distributions of real faces across different domains while imposing no compactness constraints on forged faces, thus forming a discriminative distribution characterized by "compact real and separated fake." Regarding style alignment, Wang et al. [14] proposed a Shuffle Style Assembly Network (SSAN) that separates content and style components within feature representations, creating a stylized feature space. Le et al. [15] employed a gradient alignment strategy to guide the model toward optimal flat minima regions, thereby improving generalization performance without requiring additional learning modules. Additionally, methods such as incorporating synthetic unknown attack samples into the training set—without explicitly distinguishing domains—have been developed to assist in training the feature extractor [16]. However, conventional source-domain-based generalization learning fails to utilize unlabeled data from the target domain, resulting in limited effectiveness against rapidly evolving AI-generated facial attacks.

1.2.3 Semi-supervised and unsupervised domain adaptation methods

Domain adaptation methods allow access to unlabeled data from the target domain during training, reducing distributional differences between domains through feature alignment. These methods leverage knowledge of the target domain to train models on the source domain so that they perform as well as possible on the target domain. Li et al. [17] trained a generalizable classifier by minimizing the maximum mean discrepancy (MMD) between latent features of the source and target domains. Subsequently, Jia et al. [18] proposed a unified unsupervised and semi-supervised domain adaptation network, introducing an edge distribution alignment module (MDA) and a conditional distribution alignment module (CDA) to learn a compact feature space for authentic and fake faces. This model incorporates semi-supervised learning by assuming a small amount of labeled data exists in the target domain, using this annotation information to guide the model toward more accurate adaptation to the target distribution.

Subsequent unsupervised and semi-supervised domain adaptation and domain generalization methods have increasingly incorporated various spoofing trace separation techniques as data augmentation strategies. These include using progressive feature disentanglement structures to separate liveness-related information [19], constructing causal graphs to estimate ideal causal spoofing representations [20], splitting, rotating, and recombining images with local patch embedding [21], and employing PatchMix-style data block shuffling for enhancement [22]. Such approaches further improve cross-dataset generalization performance and provide new paradigms for data augmentation in facial anti-spoofing. However, they are mostly limited to single-source semi-supervised scenarios or unsupervised settings relying on large target domain datasets.

1.3 Research Content

Current face anti-spoofing models face significant limitations in balancing generalization capability and computational efficiency, especially given the rapid evolution of facial attack methods driven by fast iterations in AI-generated models. To address the cross-domain generalization problem in face liveness detection, this paper proposes a semi-supervised one-sided adversarial domain generalization framework. Assuming that the target domain contains only a small number of labeled samples and a large number of unlabeled ones, we take multi-source domain-based one-sided adversarial domain generalization (SSDG) as the baseline model, and further incorporate a small amount of labeled target data to enhance performance.

By combining labeled domain data with a large amount of unlabeled data, and leveraging the advantages of image block content destruction for consistency-preserving data augmentation and semi-supervised lightweight data requirements, we further enhance the model's cross-domain generalization ability through target domain entropy minimization, source-target domain adversarial training, and positive sample center alignment loss. Specific innovations include:

Using SSDG[13] as the baseline model, we retain its classifier cross-entropy loss, asymmetric triplet loss, and one-sided adversarial domain discriminator for positive samples to enhance its anti-spoofing recognition capability against novel types of facial spoofing attacks.

2. Drawing inspiration from the semi-supervised module in the USDAN model [18], we propose, for the first time, a loss function tailored for multi-source domain semi-supervised scenarios, incorporating labeled target domain classification, entropy minimization, inter-domain adversarial training, and an innovative one-sided positive sample center alignment.

3. Introduce an image patch content disruption consistency loss during the early training phase on the unlabeled target domain dataset, disrupting image structures and enabling the model to focus on local textures through contrastive feature consistency loss, thereby promoting the separation of deceptive trace features and enhancing robustness against spatial structural variations.

4. Conduct ablation experiments on the classic OULU-NPU dataset in the field of facial anti-spoofing to validate the effectiveness of each component, and compare with the original SSDG and existing methods.

5. Using the OULU-NPU dataset and two newly introduced AI-generated face datasets as target domain data, we validate the framework's effectiveness across various face anti-spoofing scenarios, including novel AI-generated face liveness detection attack scenarios.

2 A Semi-Supervised Domain Generalization Framework for Face Anti-Spoofing Detection

2.1 Face Anti-Counterfeiting Detection Problem Definition

Face anti-spoofing detection, also known as liveness detection or face anti-spoofing, can be formalized as a binary classification task. Given an input image x, the model outputs a label $y \in \{0,1\}$, where 0 indicates a spoofed attack (such as printed images, video replay, 3D masks, or AI-generated faces), and 1 indicates a genuine human face. In cross-domain scenarios using semi-supervised models, the training set Ds = {(xis, yisj i=1^Ns comes from the source domain (assuming, as per conventional methods, the existence of a source domain; in this paper, multi-source domain generalization is considered, thus additional asymmetric triplet modules and one-sided domain discriminator adversarial modules are introduced to handle relationships among different source domains), with data size Ns and distribution Ps. The test set_{Dt} = {(xtj, ytj)}j=1^Nt comes *from* the target domain, with distribution Pt, where Ps ≠ Pt. The goal of domain generalization is to learn a *classifier* f: $\chi \rightarrow \{0,1\}$, where χ represents the set of feature tensors extracted from all input samples. The objective is to map any input from the unseen target domain to the output labels 0 or 1 indicating authenticity, minimizing expected error on unknown target domains.

2.2 Feature Extraction and Embedding Network

2.2.1 Image Patch Consistency Data Augmentation

Considering that effective spoofing cues in liveness detection are typically local and independent of spatial layout—such as noise patterns in printed photos, moiré artifacts from screen playback, seams at mask edges, or abnormal material reflections—these features do not rely on the overall facial shape or the positions of facial features. Even when an image is divided into small patches and randomly shuffled, these local texture characteristics remain preserved within each patch. Therefore, this model design segments images into fixed-size patches and shuffles them randomly to disrupt the global spatial structure, thereby forcing the model

The features extracted from the original and shuffled images are made as similar as possible, making the model insensitive to spatial arrangements of input images, thereby indirectly encouraging it to focus on local forgery traces while suppressing reliance on global information such as facial shape and identity.

2.2.2 Backbone network and feature embedding layer

This paper uses a ResNet18 pre-trained on ImageNet as the backbone network for spatial feature extraction. To accommodate 256×256 input images, the original fully connected layer is removed while retaining the convolutional and pooling layers. After forward propagation, the input image produces a feature map of dimension 512×8×8. The backbone network parameters are fine-tuned during training.

The feature map is fed into the feature embedding layer, and after global average pooling, a 512-dimensional vector is obtained. This vector then passes sequentially through a fully connected bottleneck layer with 512 hidden units, followed by ReLU activation and Dropout with a 0.5 dropout probability, resulting in the deep embedded feature vector *f*. Subsequently, the embedded features are L2-normalized:

$$\hat{f} = \frac{f}{\| f \|_2} \cdot \sqrt{2} \qquad \text{（式 2-1）}$$

□

Here, *f* represents the final deep embedded feature, and the model employs normalization and dropout to stabilize training and improve generalization performance.

2.2.3 Classifier

As defined in the problem statement, the classifier of the face anti-spoofing model is used to output binary classification logits, with depth □

The embedded feature *f* passes through a linear layer (512→2) classifier weights normalized by column to L2

Multiply $_{Wnorm}$ to obtain the final predicted sample label.

$$W_{norm} = \frac{W}{\| W \|_2},$$

$$\log its = \hat{f} \cdot W_{norm}^{T} \qquad \text{（式 2-2）}$$

2.3 Adversarial Learning and Gradient Reversal Layer

The general adversarial loss function for adversarial learning can be summarized as follows: let G be the feature generator and D be the domain discriminator used for training.

$$\min_{D} \max_{G} \ell_{\text{Ada}}(G, D) = -\mathrm{E}_{x, y \sim X_r, Y_D} \sum_{n=1}^{N} 1_{[n=y]} \log D(G(x)) \qquad \text{（式 2-3）}$$

where X_r and Y_D are the sample set and the corresponding domain label set, respectively, and x and y are a sampled image and its domain label from X_r and Y_D, respectively. The feature generator G *maps* the input image to a feature vector G(x), and the domain *discriminator* takes this *feature* as input and outputs a probability distribution D (G(x)) ∈ R^*N*, *representing* the probability that the *feature* belongs to each of the N domains. 1[n=y] is *an* indicator function that equals 1 when the predicted class matches the true *domain* label. The inner summation $\sum_{n=1}^{N}$ 1[n=y] log D(G(x)) represents the cross-entropy loss function, used to *compute* the true

The sum of the probabilities predicted by the discriminator for real-domain feature data under the true label. The outer layer maximizes the expectation by taking the negative sign of the sum, aiming to maximize the probability that the discriminator correctly predicts the domain labels.

In adversarial learning, the discriminator's objective is to minimize the loss and maximize the probability of correct classification.

The generator's objective is to maximize, aiming to maximize the loss and deceive the discriminator. The gradient reversal layer plays a role in the framework.

After computing the cross-entropy loss, during backpropagation for G, the gradient is reversed or the adversarial loss is negated to update G's parameters, so that the generated features reduce the discriminator's domain classification accuracy. D

During backpropagation, cross-entropy loss is normally used to increase the discriminator's domain classification accuracy; this adversarial process drives the feature generator to learn domain-invariant features.

The proposed framework incorporates two sets of domain adversarial learning, both aiming to reduce inter-domain differences but operating at different scopes. a. Unilateral adversarial learning performs adversarial training among multiple source domains, enabling the feature representations of real face samples from all source domains to become consistent in the representation space through one-sided adversarial interactions between the source domain discriminator and the feature generator. b. Semi-supervised marginal distribution alignment conducts adversarial learning between the target domain and the aggregated source domains, minimizing the marginal distribution discrepancy between the source and target domains via adversarial interactions between the unlabeled target features and the source-target domain discriminator and feature generator.

2.4 SSDG Unilateral Adversarial Domain Generalization Module

Considering that real facial images in different face liveness detection datasets are collected from actual individuals, making them easier to learn compact feature distributions, while fake facial features across different datasets are difficult to aggregate due to variations in attack types and image acquisition methods—forcing them into a single category may lead to model confusion—in 2020, Jia Y. et al. proposed an end-to-end single-source domain generalization framework (SSDG) [13]. This model treats fake faces from different domains as distinct classes to disperse their representations, while grouping all real faces across domains into a single class to cluster them together. By applying different optimization strategies for real and fake data, the model achieves clearer generalization boundaries between classes.

Its specific architecture is shown in Figure 2-1, where input faces with different colored borders represent examples from different domains. The fake face image data and real face image data from all domains are separately fed into two feature generators—fake face and real face—that share parameters. After training and L2 normalization, the real face recognition data undergoes single-face adversarial learning between the domain discriminator and the real face feature generator, enabling the feature distributions across different domains to align with those of the real data.

Faces are indistinguishable, while fake face data are differentiated without adversarial processing. Furthermore, to better exploit asymmetric triplet mining for separating fake faces and simultaneously aggregate real facial features across different domains, the model encourages fake face features to be more dispersed in the feature space. The entire model incorporates feature and weight normalization to further enhance generalization capability.

This section of the framework design discusses only multiple source domain sample data and does not include labeled samples from the target domain.

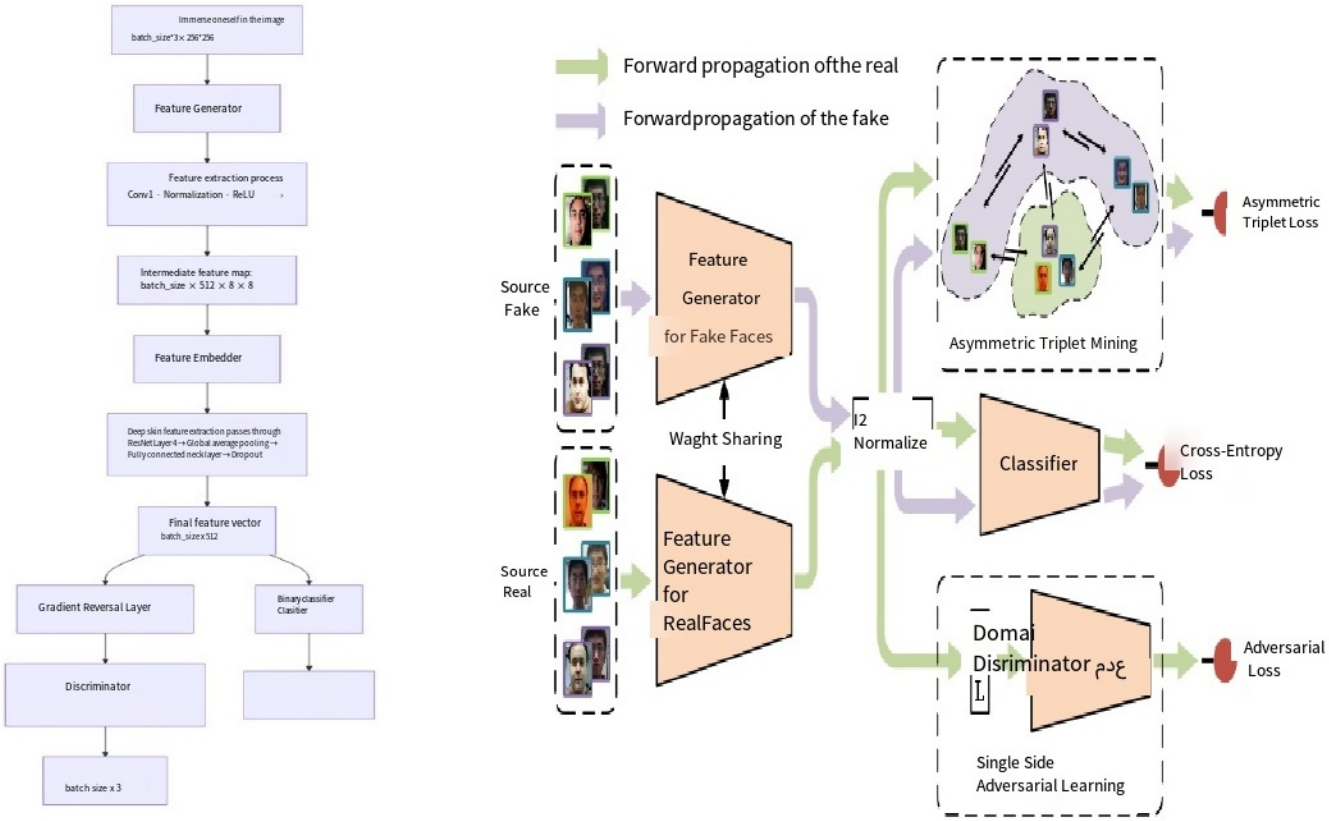


Figure 2-1 Model architecture of the end-to-end single-source domain generalization framework (SSDG)

2.4.1 Unilateral adversarial learning

This stage proposes one-sided adversarial learning to learn a multi-source domain-invariant feature space that acts solely on authentic facial features, while fake faces do not participate in the adversarial process. The domain discriminator operates within the range of real samples to determine which source domain the input features originate from, and interacts with the feature generator exclusively to enhance the quality of real samples.

This is a one-sided adversarial learning that targets domain-invariant features while being ineffective against fake samples. The learned feature generator parameters are shared across real and fake samples in the source domain, as well as all samples in the target domain.

2.4.2 Asymmetric Triple

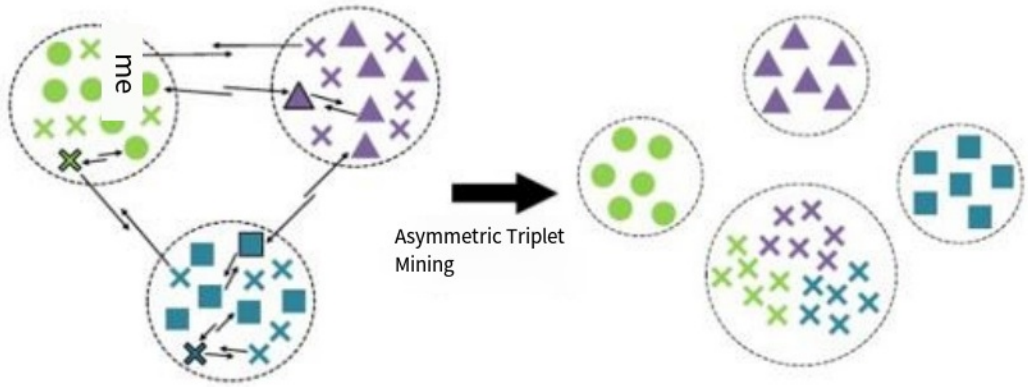


Figure 2-2 Schematic of an asymmetric triplet

Asymmetric triplet mining is employed within the SSDG framework to optimize the feature generator. In this setup, all real faces from the source domains are treated as a single class (sharing the same label), while fake faces from each of the three distinct source domains are considered independent classes (with different labels). The optimization objectives include "aggregating multi-source real faces," "clearly distinguishing real from fake faces," and "classifying fake faces from different source domains," enabling the model to preserve and learn diverse differences.

Its loss function is given by the following formula:

$$\min_{G} \ell_{\text{ASTrip}}(G) = \sum_{x_q^a, x_p^b, x_n^c} (\| f(x_q^a) - f(x_p^b) \|_2^2 - \| f(x_q^a) - f(x_n^c) \|_2^2 + \alpha) \qquad \text{（式 2-4）}$$

For an anchor sample $_{xqa}$, samples with the same label are designated as positive samples *xbp*, *and those with* different labels are considered negative samples xnc. The $_{model}^{expects}$ that the squared Euclidean distance between the anchor feature and the positive sample feature should be at least α smaller than the squared distance to the negative sample feature. If this condition is not met, the model incurs a positive loss, and the feature generator is updated via backpropagation to reduce the distance difference.

Each triplet corresponds to one positive and one negative sample. To achieve asymmetric four-class classification based on positive and negative samples from three source domains, a large number of triplets of various types are constructed during training. The sum of their losses is computed, covering all possible combinations so that the feature generator learns the desired feature space arrangement.

2.5 Semi-supervised domain adaptation training framework

2.5.1 Edge Distribution Alignment

Due to the diversity of data collection conditions, there are significant differences in the marginal distributions of features between the source and target domains. The core idea of this module is to minimize the marginal distribution discrepancy between the source and target domains through adversarial domain adaptation. In this model, only unlabeled samples from the target domain are used together with features from multiple source domains for adversarial training, while the domain-invariant feature parameters obtained still remain shared across all training data.

2.5.2 Principle of Minimum Entropy

Adding an entropy minimization loss to the training process for unlabeled target domain samples aims to bias the model's predictions toward low cross-entropy, highly concentrated probability distributions, thereby facilitating the utilization of unlabeled data. For a probability distribution *p = [p*□, p□, …, p_c*]*, the Shannon entropy is defined as:

$$H(p) = -\sum_{c=1}^{C} p_c \log p_c \qquad \text{（式 2-5）}$$

The sample label classification prediction result is a probability vector indicating the likelihood of the sample belonging to each class. A smaller entropy value signifies significant differences among class probabilities, a more concentrated distribution (closer to one-hot), and thus greater model confidence. In semi-supervised learning, applying an entropy minimization loss to unlabeled samples forces the model's decision boundary to pass through low-density regions, thereby improving classification performance. This paper calculates the entropy loss based on the predicted probabilities for unlabeled target domain samples:

$$\ell_{ent} = -\frac{1}{N_u}\sum_{i=1}^{N_u}\sum_{c=1}^{2} p_{i,c}\log p_c \qquad \text{（式 2-6）}$$

2.5.3 Improve positive sample center alignment

To align the feature centers of同类 samples from the source and target domains, a center alignment module is designed based on domain-invariant features generated from the source and target domains. This paper performs center alignment only on labeled real samples from both the source and target domains to avoid interfering with the domain separation characteristics of forged samples. Let $\{_{fis}\}$i=1N be the set of feature vectors for real source domain samples, and $\{fjt_{j=1N}$ be the set of feature vectors for real $_{target}$ domain samples, with $_{their\ respective\ feature}$ centers defined as:

$$\mu_s^r = \frac{1}{N_s^r}\sum_{i=1}^{N_s^r} f_i^s, \mu_t^r = \frac{1}{N_t^r}\sum_{j=1}^{N_t^r} f_j^t \qquad \text{（式 2-7）}$$

The center alignment loss is the Euclidean distance between two centers:

$$\ell_{center} = \| \mu_s^r - \mu_t^r \|_2 \qquad \text{（式 2-8）}$$

3 Model Design and Algorithm Implementation

3.1 Model Framework Overview

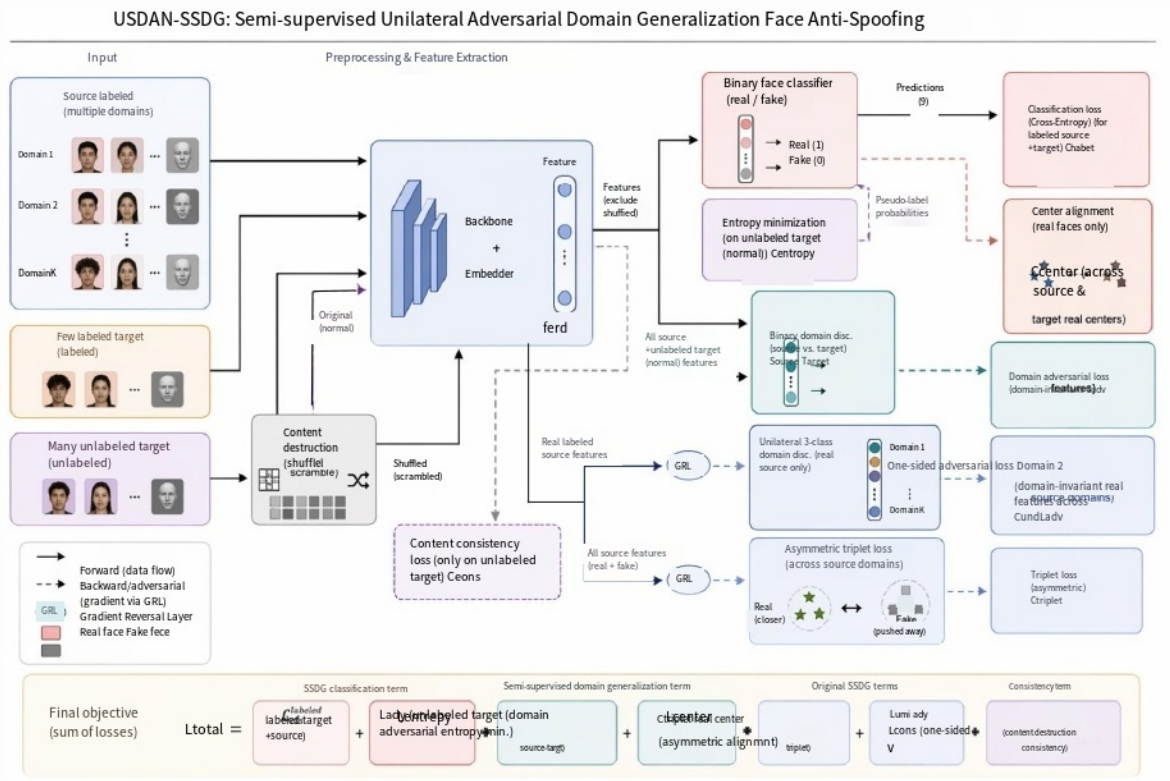


Figure 3-1 USDAN-SSDG Model Framework Diagram

The overall architecture of the proposed semi-supervised one-sided adversarial domain generalization facial liveness detection model (USDAN-SSDG) is shown in Figure 3-1. The model consists of four main components: data augmentation and feature extraction backbone, embedding network, classifier for authentic/fake face discrimination, and two domain discriminators—a one-sided domain ternary adversarial discriminator and a binary classification adversarial domain discriminator. The one-sided domain discriminator is used exclusively for source domain classification and adversarial confusion on samples with true labels, while the binary domain discriminator performs cross-domain adversarial confusion and marginal distribution alignment between all extracted features from the three source domains and those from the target domain.

During training, the model simultaneously receives labeled samples from multiple source domains, a small number of labeled samples from the target domain, and a large number of unlabeled samples from the target domain. The labeled samples from both the multi-source and target domains are directly fed into the feature extractor and embedding network, while the unlabeled samples from the target domain first undergo image patch content consistency disruption to generate shuffled structural samples through content corruption. These original and shuffled samples are then jointly input into the feature extraction and embedding network.

After the embedder outputs the embedding features, the shuffled features of unlabeled samples are used solely for computing content corruption consistency loss by comparing them with normal sample features. Subsequently, all features except those from shuffled unlabeled target domain samples are fed into a binary classifier to predict whether they belong to real or fake faces. Then, labeled and unlabeled data are processed separately: (1) For unlabeled normal target domain samples, confidence entropy loss is calculated based on the predicted probabilities of being real or fake, and their features are concatenated with source domain features and input into a domain discriminator to perform domain classification and compute domain adversarial loss, thereby learning cross-target-domain-invariant features. (2) For labeled source and target domain data, classification cross-entropy loss is computed by comparing predicted labels with ground-truth labels. Features labeled as real faces participate in computing center alignment loss between the source and target domains, pulling closer the feature centroids of genuine face samples across domains. This completes the semi-supervised processing of the target domain. The source domain features then proceed through two branches to fulfill the core SSDG training framework: ① Embedding features from the source domain with true labels are fed into a one-sided adversarial domain discriminator to determine which source domain they belong to; after passing through a gradient reversal layer (GRL), one-sided adversarial loss is computed, forcing real samples to learn domain-invariant features across three source domains. ② All source domain samples' features are used in an asymmetric triplet loss computation, further reducing the inter-domain distances among real samples from multiple sources while distancing the centroids of fake samples from different source domains.

The final loss function of the framework consists of: ① the classification loss, asymmetric triplet loss, and one-sided adversarial loss for real samples from the original SSDG loss; ② the labeled target domain classification loss and unlabeled...

The loss function is composed of a weighted sum of the following components: ① label prediction entropy minimization loss for labeled target domain; ② source-target domain feature adversarial loss and positive sample center alignment loss for labeled target domain; and ③ consistency loss based on data augmentation that disrupts the content of unlabeled target domain samples.

The total loss function is given by:

$$\ell_{total} = \ell_{cls} + \lambda_t \ell_{triplet} + \lambda_a \ell_{real_adv} + \lambda_l \ell_{labeled_tgt_cls} + \lambda_e \ell_{ent} + \lambda_d \ell_{transfer} + \lambda_c \ell_{center} + \lambda_{con} \ell_{MSE} \quad \text{（式 3-1）}$$

3.2 Dataset Introduction

To align with the semi-supervised domain generalization framework for face anti-spoofing presented in this paper, multiple databases are required to conduct experiments. The initial experimental model and ablation studies are built upon four of the most classic and widely used public benchmark datasets in the field of Face Anti-Spoofing (FAS): CASIA-FASD (C), Idiap Replay-Attack (I), MSU-MFSD (M), and OULU-NPU (O). These four datasets differ significantly in terms of acquisition devices, lighting conditions, attack types, and subject ethnic backgrounds, making them suitable for cross-database testing to evaluate the model's domain generalization capability. Since our model does not require temporal cue analysis, a lightweight training approach is adopted by extracting only a few frames from each video during preprocessing across all four datasets. In this study, the preprocessed and reduced versions of datasets C, I, and M serve as source domains (each fully labeled), while dataset O acts as the target domain. Only a portion of its training data is used—20% labeled and 80% unlabeled—as labeled/unlabeled semi-supervised samples for training. Validation and testing still follow the official splits provided by dataset O.

During the generalization capability validation phase, considering new types of AI attacks, two AI-generated face detection datasets from Kaggle were incorporated. After dividing the data into training (50%), validation (25%), and test sets (25%), each AI dataset was combined with a small portion of the O dataset for training.

The validation set is concatenated, and the resulting fused training set undergoes semi-supervised training with a ratio of 20% labeled and 80% unlabeled data. The two newly trained OULU_AI fusion models are used to predict on the OULU test set and AI test set, respectively, to evaluate the models' generalization performance on real attacks and synthetic faces. Detailed descriptions of each dataset are shown in Table 3-1.

Table 3-1 Introduction to the Experimental Dataset

	Dataset	Source Institution	Sample composition	Collection device	Attack/Forgery Type	This article uses settings (number of images/frames)
Source	CASIA-FASD (C)	Chinese Academy of Sciences, Institute of Automation,	50 Chinese people (Laptop,	USB	Bending paper printing, vision	Approximately 3,200 frames (one frame per video) used
Domain		The	Subject	Infrared camera	Frequent playback, crop eye area	Training
	Idiap Replay-Attack (I)	Swiss Idiap Research	50 European	Built-in laptop camera	Photo printing, screen refresh	1200 keyframes used for training, real and fake faces
		The	Subject	Like a head	Place (handheld/fixed)	Each half
	MSU-MFSD (M)	Michigan State University	35 different types	MacBook Air front	Paper print, screen refresh	1,400 keyframes used for training, real and fake faces
		Study	Subject of the tribe	Placement, Nexus 5 back	Release (indoor and outdoor versions available)	Each half
				Place	Resolution	
Eye	OULU-NPU	University of Oulu, Finland	55 names of various types	Six smartphone models	Print/Video Playback +	Train 1,500 images (300 labeled + 1,200 unlabeled)
Label		Northern China Industrial	Subject of the tribe	Front-end	Lighting and Background Changes	Zhang (used for semi-supervised training), validation
Domain		Industry University				1,000 pieces, test 1,000 pieces
	AI-Generated Face	Kaggle	Different races	—	AI-generated faces (GAN)	Training 1601 images (320 labeled + 1281
	Detection		with gender		etc.	No labels, validated 802, tested 800
	Dataset(AI_I)		Facial structure			Zhang
	AI-Human Faces	Kaggle	Pixel-level fusion	—	AI-generated faces (GAN/	Train 4,815 images (20% labeled), validate
	Dataset(AI_II)		Adult face		Diffusion models	2,408 certificates and 2,407 tests

3.2.1 Introduction to the New Target Domain Dataset

a. AI-Generated Face Detection Dataset (Kaggle)

This dataset consists of high-quality real facial images and AI-generated synthetic facial images, specifically designed for machine learning and deep learning classification tasks. The dataset contains 2,202 real samples and 1,001 AI-generated samples, with the latter intentionally covering diverse racial and gender-based facial structures. Although the number of AI-generated samples is relatively small, their high diversity makes the dataset well-suited for testing model performance on small sample sizes.

The discriminative performance on this synthetic face dataset. Before model training, the dataset was divided into 1,601 training images (320 labeled and 1,281 unlabeled), 802 validation images, and 800 test images. The data was randomly shuffled in sequence prior to partitioning to ensure that the proportions of real and fake faces in each subset were consistent with those in the original dataset. This dataset is referred to as AI_I in the following text.

b. AI-Human Faces Dataset (Kaggle)

This dataset contains approximately 9,600 facial images, divided into two subsets: real faces and AI-generated faces. It includes 5,000 real images and 4,630 AI-generated images. The AI-generated images are produced by generative adversarial networks (GANs) or diffusion models, making them suitable for deepfake detection and image authenticity verification tasks. The dataset exhibits significant distributional differences from traditional source-domain attack types—where the source domain involves low-quality print/replay attacks, while the target domain consists of pixel-level synthetic faces—effectively testing a model's transferability against novel facial attack methods. Prior to model training, the dataset is split into a training set of 4,815 images (20% labeled, 80% unlabeled).

,

2,408 validation images and 2,407 test images, hereinafter referred to as the AI_II dataset.

3.3 Environment Configuration and Experimental Setup

3.3.1 Hardware and Software Environment

During the experimental dataset preprocessing stage, to meet the compatibility requirements for face detection and cropping using MTCNN, an independent environment with PyTorch 0.4.1, MXNet 1.6.0, and MTCNN was employed.

All other experimental stages were run on a single NVIDIA GeForce RTX 3060 GPU (6GB VRAM), using PyTorch 2.1.0 + CUDA 12.1 and Python 3.8. To eliminate the impact of randomness, all experiments fixed the random seed to 888 and set torch.backends.cudnn.deterministic=True.

3.3.2 Data Preprocessing and Labeling

All input videos were processed to extract image datasets using uniformly sampled frames, selecting a small number of frames at regular intervals. Face detection and alignment were performed using the MTCNN technique, normalizing detected faces to 256×256×3 (RGB three channels) as network inputs. A random gradient descent optimizer was employed, selecting the face with the highest confidence score, cropping it, and returning the aligned face. The face was then uniformly scaled to 256×256 pixels to complete preprocessing. The comparison before and after processing is shown in Figure 3-2.

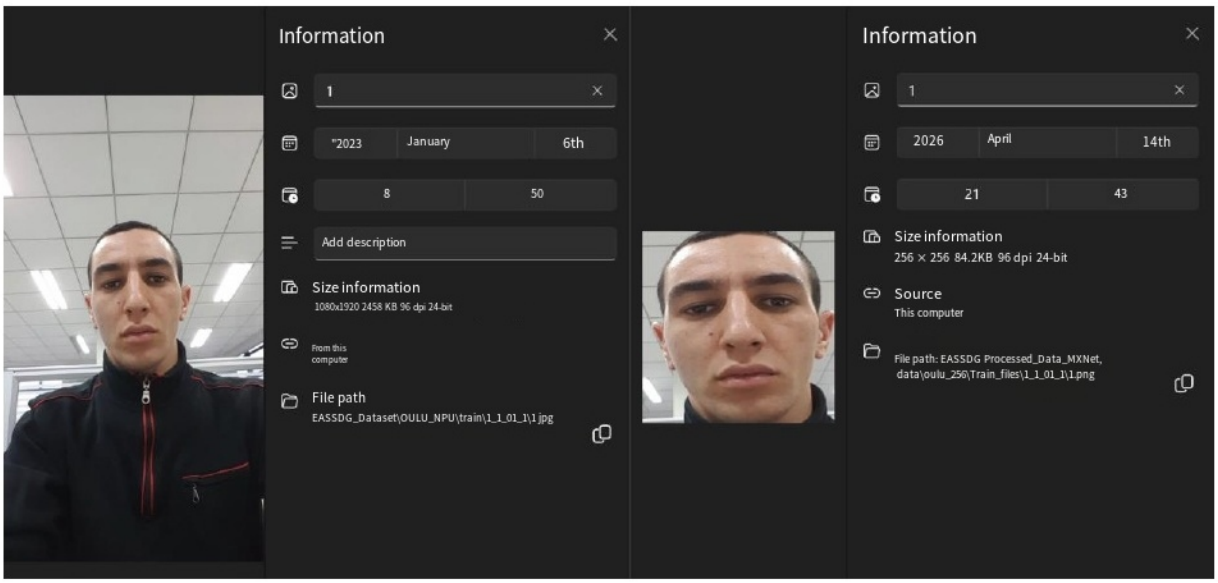


Figure 3-2 Comparison of images processed by MTCNN technology on the sequential dataset

Next, the dataset is divided. For the preprocessed image data, keywords are extracted from filenames or paths according to the naming conventions of each dataset, and samples are categorized into real/fake labels, then recorded in the corresponding training/test/validation JSON files. Specifically, for the MSU dataset, the official split provided in train_sub_list.txt and test_sub_list.txt is used: the video folder names determine whether a sample belongs to the training or test set, while "real" or "attack" in the path indicates the real/fake label. For the CASIA and Replay datasets, labels and completion types are determined based on the file paths. For the Oulu dataset, the label (d) is determined by the last digit in the folder name, and the completion type is determined from the path.

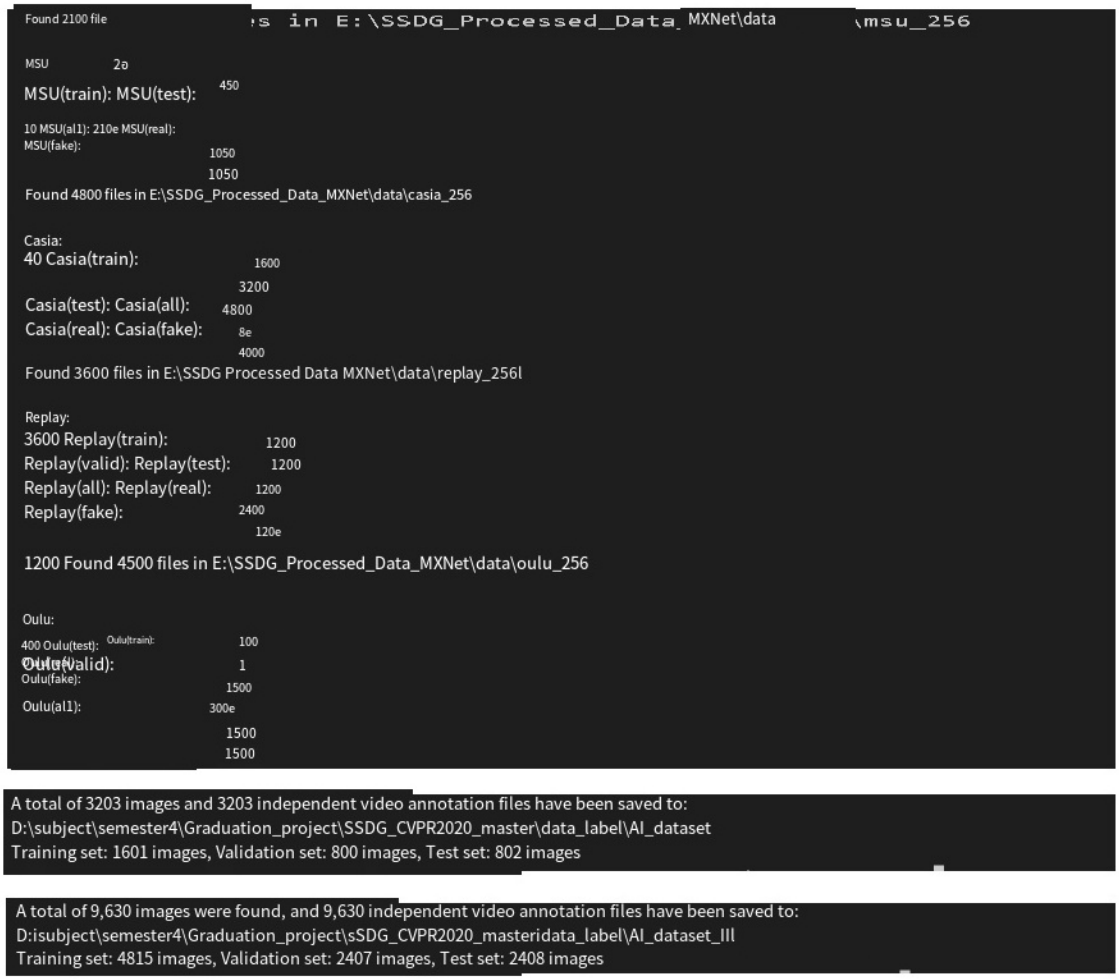


Figure 3-3 Illustration of the dataset labeling process (the actual training, validation, and testing datasets are uniformly selected from the above labeled dataset; therefore, the number of samples used for model training and validation is less than the total number of labeled samples)

During training, random horizontal flipping (with a probability of 0.5) is applied as initial data augmentation, followed by normalization to ImageNet statistics (mean [0.485, 0.456, 0.406], standard deviation $[0.229,\ 0.224,\ 0.225]$) after converting to tensors. During validation and testing, only normalization is performed without any random augmentations.

Considering that images from different datasets often contain domain-specific factors such as lighting and background, these interfering factors may lead FAS models to generate incorrect correlations, hindering robust predictions based on intrinsic spoofing feature cues. To address this issue, this paper draws inspiration from the unsupervised domain adaptation-based SDA-FAS model, extracting a portion of unlabeled target domain data during semi-supervised input. During the first 90 training iterations, images are divided into 4×4 small patches and randomly shuffled to disrupt local spatial structures.

We construct the framework and compute the content consistency MSE loss between the features generated from the shuffled image and those from the original image, thereby reducing the model's feature generator's sensitivity to the overall spatial arrangement of input images during early training stages. This enables the model to quickly focus on local features that are more prone to forgery traces, achieving domain alignment.

3.3.3 Hyperparameter Configuration

The training process uses the stochastic gradient descent (SGD) optimizer with a momentum of 0.9 and weight decay of 5×10^{-4}. The initial learning rate is set to 0.01, which is decayed by a factor of 0.1 at the 10th and 20th epochs. The maximum number of iterations is 100 epochs (each epoch consists of 10 iterations, totaling 1,000 iterations). The batch size is set to 10. During forward propagation, since six source domain sub-batches and two target domain sub-batches are concatenated, approximately 80 images are processed in each iteration.

The hyperparameters in the loss function are set as follows:

Triplet loss weight λ_triplet = 2, one-sided adversarial loss weight λ_adversarial = 0.1, labeled target domain classification loss weight λ_labeled_tgt = 0.5; entropy minimization loss weight λ_entropy = 0.1; domain adversarial loss weight λ_transfer = 1.0; center alignment loss weight λ_center = 0.05; consistency loss weight λ_consistency = 0.001; feature content consistency segmentation (shuffled) patch size patch_size = 4

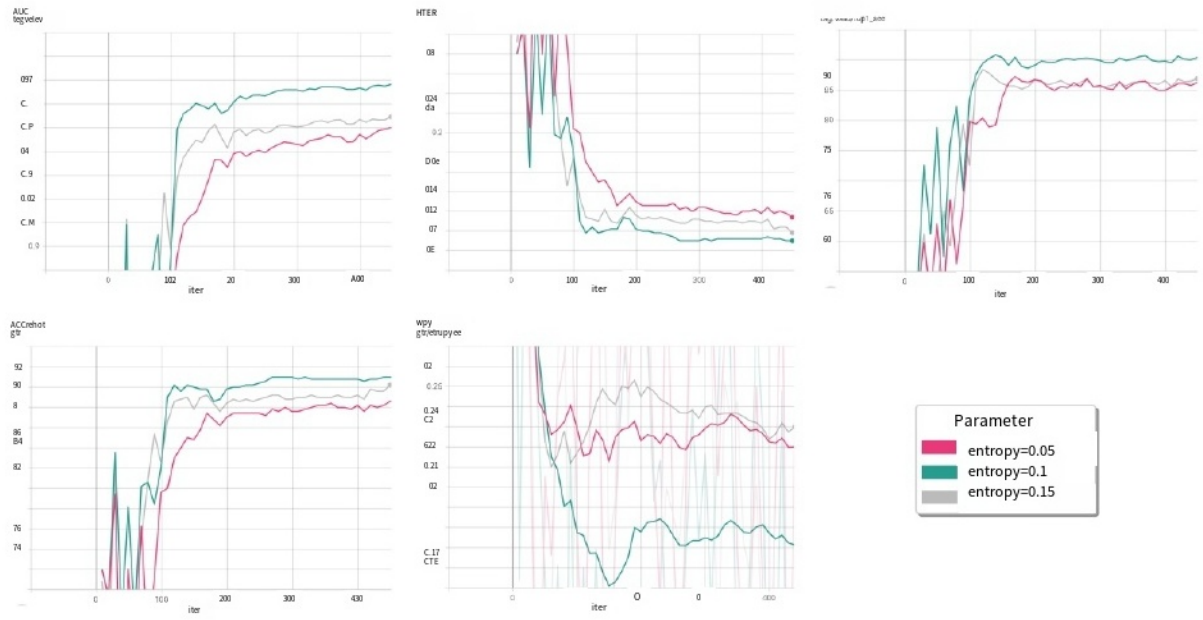


Figure 3-4 Comparison of validation set performance of the proposed model under the condition of modifying only the entropy minimization loss weight (λ_entropy)

The loss weights in the hyperparameters were determined by training the model with one variable at a time across a set of candidate weight combinations, using only the OULU dataset as the target dataset. As shown in Figure 3-4, while keeping all other parameters at their standard configurations, we varied the entropy minimization loss weight for the unlabeled data in the target domain to 0.05, 0.1, and 0.15, respectively. It was found that after 450 training iterations, the validation set achieved optimal performance in terms of ACC, HTER, and AUC curves when λ_entropy = 0.1, and the training loss converged fastest under this setting. Therefore, λ_entropy = 0.1 was selected as the standard configuration for the entropy minimization loss weight. In a similar manner, each hyperparameter combination was evaluated individually, and the one yielding the best performance on the validation set was chosen as the final hyperparameter configuration.

3.4 Evaluation Criteria

To comprehensively evaluate model performance, the following five metrics are used:

a. Accuracy (ACC@0.5): The classification accuracy calculated by binarizing the model's predicted positive class probabilities using a fixed threshold of 0.5.

b. Half Total Error Rate (HTER): At the threshold corresponding to Equal Error Rate (EER), compute the average of False Acceptance Rate (FAR) and False Rejection Rate (FRR).

$$HTER = \frac{FAR + FFR}{2} \quad \text{（式 3-2）}$$

where FAR = FP/(FP+TN) and FRR = FN/(FN+TP). A lower HTER indicates that the model has a better balance in handling both types of errors.

c. Equal Error Rate (EER): The error rate when the threshold is adjusted such that the False Acceptance Rate (FAR) equals the False Rejection Rate (FRR), reflecting the model's inherent discriminative ability.

d. Area under the ROC curve (AUC): The ROC curve plots false positive rate on the x-axis and true positive rate on the y-axis. AUC represents the area under this curve, ranging from 0.5 to 1, with higher values indicating stronger ranking performance.

e. Best threshold accuracy (ACC@thr): Search for the threshold that maximizes accuracy on the validation set, and use this threshold to compute accuracy on the test set.

3.5 Ablation Analysis

3.5.1 Semi-supervised module ablation study

To validate the effectiveness of each semi-supervised component, seven sets of control experiments were primarily designed:

Baseline: The original SSDG model, using only source domain supervised loss (classification + triplet + one-sided adversarial).

L: Add labeled target domain classification loss (λ_labeled_tgt=0.5) on top of the Baseline

。

L+E: Add unlabeled target domain entropy minimization loss (λ_entropy=0.1) on top of L.

L+T: Add domain adversarial loss on top of L (λ_transfer=1.0).

L+C: Add center alignment loss (λ_center=0.05) on top of L.

L+T+C: All semi-supervised groups remove the entropy minimization loss.

Full: Includes all of the above semi-supervised groups.

All experiments used the same random seed, hyperparameters, and number of training iterations, and were evaluated on the same validation set (OULU). We recorded the evaluation metrics on the validation set after 999 iterations (i.e., the 100th epoch), and the ablation study results are shown in Table 3-2. From the validation set results, we can observe:

Table 3-2 Quantitative results comparison of the ablation experiments on the validation set at 999 iterations

Model	ACC_threshold (%)	AUC (%)	EER (%)	HTER (%)	loss	top1_acc (%)
(baseline)SSDG_baseline	74	83.37	26	26	0.993	73.2
(L)SSDG+labeled	89.2	96.3	10.8	10.8	0.3287	88.4
(L+T)SSDG+labeled+transfer	92.4	97.32	7.6	7.6	0.3318	91.8
(L+E)SSDG+labeled+entropy	88.8	96.68	11.2	11.2	0.3801	88.8
(L+C)SSDG+labeled+center	89.6	96.9	10.4	10.4	0.4083	89.8
(L+T+C) without entropy	92.8	98.41	7.2	7.2	0.2667	92.8
(full)SSDG+USDAN_all	93.6	98.6	6.4	6.4	0.2359	93.8

1. Labeled target domain classification significantly contributes to improving the model's detection capability on the target domain. Compared to the baseline, simply incorporating the labeled target domain classification loss (L) reduces the HTER from 26.0% to 10.8%, and increases the AUC by nearly 13 percentage points. This indicates that a small number of target domain labels (20% of the target domain training images) can effectively guide the model to adapt to the target domain distribution, serving as the primary source of performance improvement.

2. Source and target domain adversarial training (L+T) effectively enhances the model's robustness in distinguishing authentic from fake faces in the target domain. After incorporating domain adversarial loss on top of L, the HTER further decreases to 7.6%, while AUC

Rising to 97.32%. This demonstrates that aligning the feature distributions between the source domain and the unlabeled target domain through gradient reversal layers effectively reduces domain discrepancies, facilitating knowledge transfer.

3. Although label-free entropy minimization (L+E) alone has relatively weak effects, it still improves both AUC and ACC accuracy at a fixed threshold of 0.5 compared to using only labeled data for classification. The HTER of L+E is 11.2%, slightly higher than that of L (10.8%), possibly because the entropy loss introduces some noise in the early training stages by constraining the confidence of predictions on unlabeled target domains. Nevertheless, overall performance remains better than the baseline and within an acceptable range. To verify whether unlabeled classification entropy minimization effectively contributes to the overall model, we additionally trained a model excluding this component (L+T+C). Results show that this model performs worse across all metrics compared to the full model, further confirming that the entropy loss works synergistically with other components to deliver positive effects in the complete model.

4. Center alignment (L+C) between labeled target domain real data and source domain real data helps stably improve the model's generalization ability, even when only incorporating the labeled target domain classification loss. Specifically, L+C reduces HTER to 10.4% and increases AUC to 96.90%, indicating that aligning class centers between the source and labeled target domains contributes to building a more compact feature space.

5. The full model achieves optimal performance: after incorporating all semi-supervised components, the HTER drops to 6.4%, and the AUC reaches 98.60%, representing reductions of 19.6 percentage points and improvements of 15.2 percentage points compared to the baseline, respectively. The validation loss also decreases from 0.993 to 0.236, indicating good model convergence.

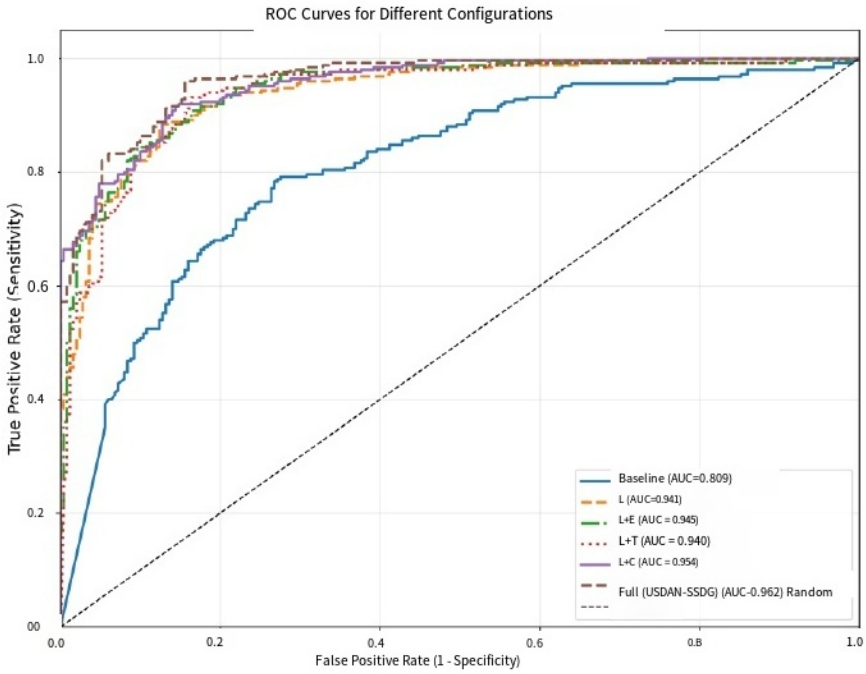


Figure 3-5 ROC curves of semi-supervised ablation models on the OULU test set

3.5.2 Image Consistency Data Augmentation Ablation Study

To independently verify the effectiveness of the image consistency data augmentation module, this paper first designs an experiment in which all unlabeled images are subjected to image consistency checks during the entire 1000-iteration semi-supervised training process of the overall model on SSDG+ Oulu dataset. The corresponding loss is then incorporated into backpropagation. As shown in Figure 3-6, the image consistency loss converges around iterations 90−100, after which it no longer contributes meaningfully to model training. Therefore, we choose to apply the image consistency loss only during the first 90 iterations to encourage the feature generator to focus on local features that remain unaffected by image scrambling blocks.

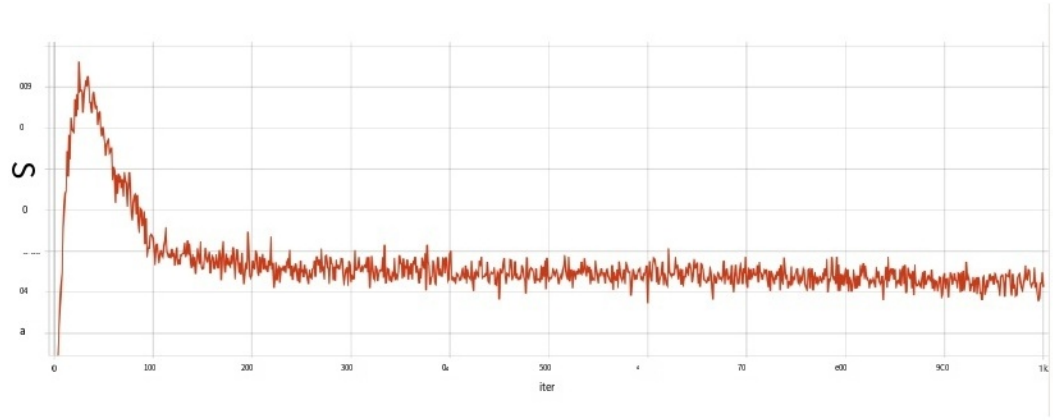


Figure 3-6: The decrease in unlabeled data image consistency loss during 1,000 iterations of semi-supervised training on the Oulu dataset

All experiments used the same random seed, hyperparameters, and number of training iterations, and were evaluated on the same test set (OULU). The comparison results are shown in Table 3-3:

Table 3-3 Comparison of model testing performance before and after incorporating unlabeled data from the target domain in the semi-supervised pre-training stage with image shuffle block consistency enhancement

Model	Test acc (%)	ACC_threshold (%)	HTER (%)	AUC (%)
USDAN_SSDG(without shuffle)	89.4	87.6	12.4	96.01
USDAN_SSDG(all shuffle)	77.4	82.8	17.2	91.39
USDAN_SSDG(9% iter shuffle)	90.4	90.80	9.20	96.89

Observations show that incorporating image patch consistency augmentation throughout the entire training process can, due to the addition of unnecessary consistency loss in later stages, interfere with the model's normal generalization training and lead to a decline in test performance. However, applying image patch consistency data augmentation only during the early training phase can further improve the model's generalization capability.

3.6 Feature visualization results

To intuitively understand the impact of semi-supervised components on the feature space, t-SNE was used to reduce 512-dimensional features to a 2D plane. Randomly selected 500 samples from both the source and target domains (250 per class), and features were extracted separately using the Baseline model, the model without image shuffle block consistency data augmentation, and the complete model. The results are shown in Figures 3-7, 3-8, and 3-9.

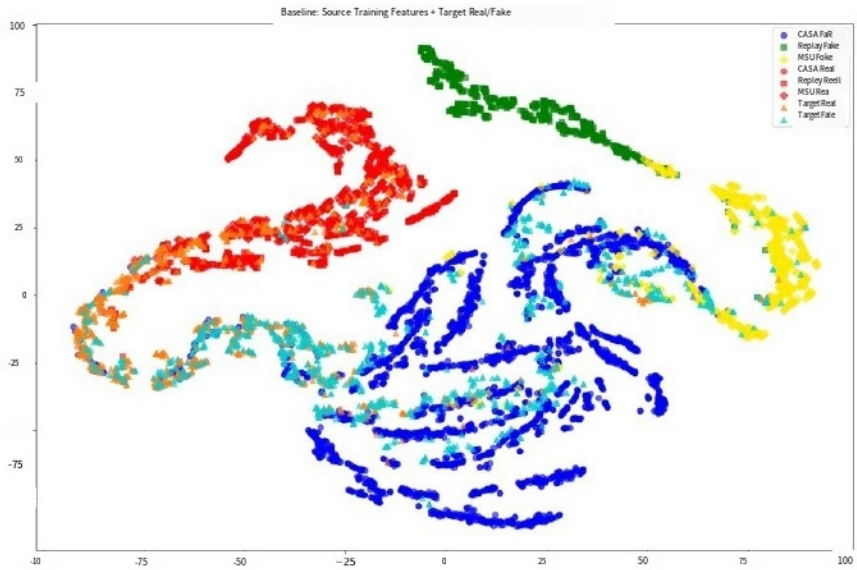


Figure 3-7 Baseline model source domain target domain real and fake facial features

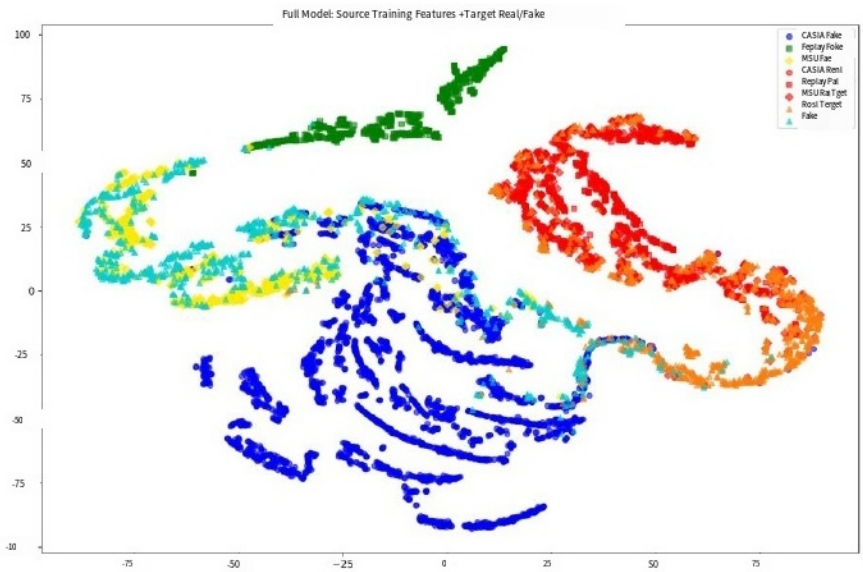


Figure 3-8 Source and target domain real and fake facial features of the model without image scrambling block consistency enhancement

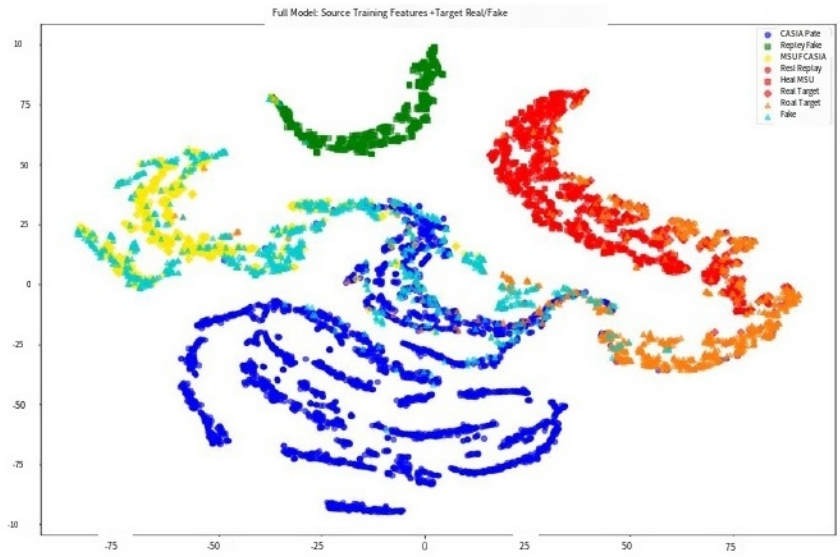


Figure 3-9 Final model source domain and target domain real and fake facial features

Observation reveals:

In the baseline model, there is significant overlap between real and forged samples in the target domain, with target domain samples scattered around the source domain without clear clustering. Additionally, the feature distributions of the fake labels across the three source domains show noticeable adhesion.

b. In the model without consistency enhancement after incorporating semi-supervised optimization, there is still overlap between source domain MSU and CASIA forged samples, while partial overlap exists between authentic and forged samples in the target domain, with no clear distinction observed between genuine and fake samples in the target domain.

In the complete model, real and fake samples from the target domain form two distinct clusters. Real samples across all domains are clearly grouped together, while the intra-cluster distances among fake samples from each source domain significantly decrease and the inter-cluster distances increase substantially. Meanwhile, samples from the target domain and source domains are highly mixed within the same class, indicating that domain differences have been effectively alleviated.

4 Experimental Results and Analysis

4.1 Result Analysis

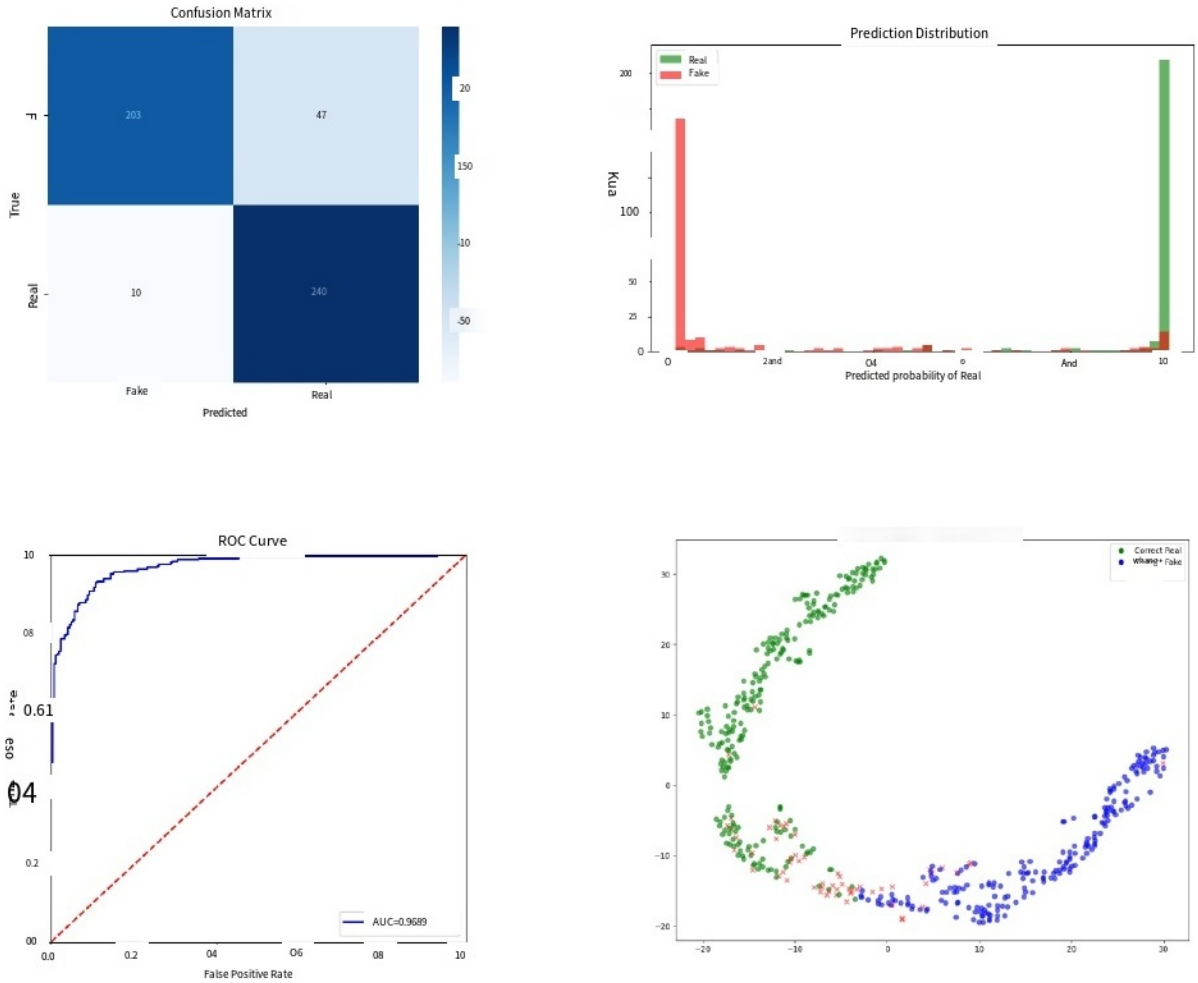


Figure 4-1 Analysis of USDAN-SSDG model on the test set (top left: confusion matrix with video as the average detection unit; top right: predicted probability distribution; bottom left: ROC curve; bottom right: true and false features of real and fake faces predicted on the test set)

Distribution map

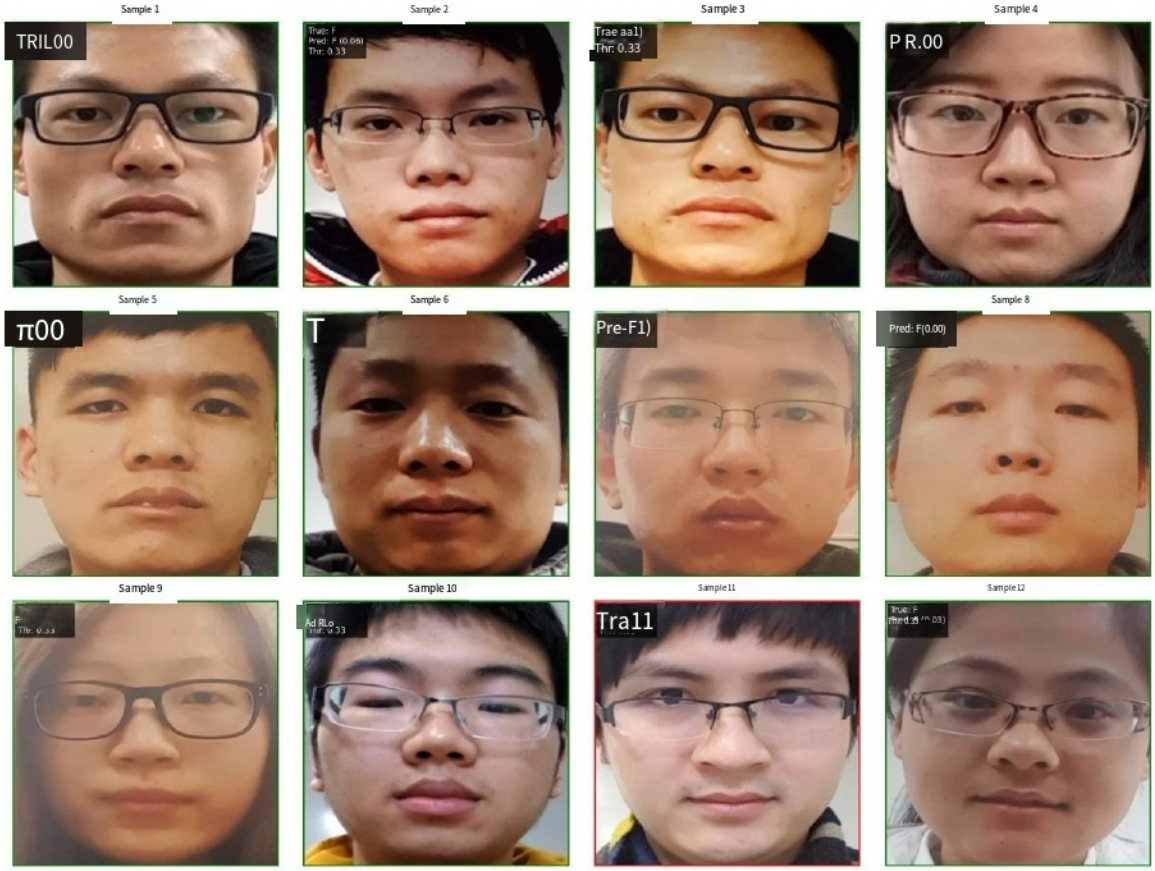


Figure 4-2 Model prediction results on the test set (green boxes indicate correct predictions, red boxes indicate incorrect predictions), labeled
Predicted labels, predicted probabilities, and optimal model threshold

The model's performance was evaluated on a test set of 1,000 frames (two frames per video) from 500 OULU videos. As shown in the figure above, the model demonstrates strong discrimination between real and fake faces, with prediction errors primarily occurring at regions where authentic and synthetic features are easily confused.

4.2 Comparison with existing models

To validate the advancement and limitations of the proposed method, it is compared with several existing liveness detection methods on the simplified OULU test set used in this paper. The comparison methods include:

Original SSDG-ResNet18 (SSDG_R): Uses only source domain supervision loss, with no semi-supervised components.

DGUA_FAS[16]: A state-of-the-art domain generalization liveness detection method based on incorporating a generator for simulating unknown features.

IADG[23]: A Style-Aligned Perception-Based Instance Domain Generalization Liveness Detection Method Without Domain Labels.

Table 4-1 Evaluation Metrics for Multi-source Domain Cross-dataset Testing

Model	Test acc (%)	ACC_threshold (%)	HTER (%)	AUC (%)
USDAN_SSDG (This article)	90.4	90.80	9.20	96.89
SSDG (R)	75.8	74.80	25.2	80.91
DGUA_FAS[16]	75.4	90.00	10.00	97.08
IADG[23]	69.70	75.90	24.10	82.62

From the data in Table 4-1, it can be seen that:

Our method significantly outperforms the original SSDG: our approach (SSDG+USDAN) reduces HTER from 25.2% to 9.2%, a decrease of over 16 percentage points, and improves AUC by 16.0 percentage points. This clearly demonstrates the effectiveness of the semi-supervised components—labeled classification, entropy minimization, domain adversarial training, and center alignment—as well as the advantage of collaborative learning between multi-source domains and unlabeled target domain data.

2. Compared to DGUA_FAS, the proposed method achieves better performance on the core metric HTER (9.2% vs. 10.0%), with a slightly lower AUC by 0.19 percentage points, but a significantly higher accuracy of 15.0 percentage points at the default threshold. The best threshold accuracy of DGUA_FAS (90.0%) is notably higher than its default accuracy (75.4%), indicating that its model output probabilities are somewhat biased and require fine-tuning of the threshold to achieve optimal performance; in contrast, the default accuracy of the proposed method is nearly equal to its best threshold accuracy.

(90.4% vs. 90.8%)

indicates better model calibration, which is more favorable for practical deployment. At the same time, compared to

The simulation of unknown features in the DGUA model, along with USDAN_SSDG's direct use of unlabeled data from the target domain achieving better AUC results, further demonstrates the design advantage of incorporating unlabeled data within a semi-supervised framework.

3. Meanwhile, this model significantly outperforms the IADG passive domain label-based style alignment domain generalization anti-spoofing model in detection performance, to some extent demonstrating the substantial lightweight advantage of the multi-source domain semi-supervised one-sided adversarial domain generalization strategy compared to complex universal domain adaptation methods.

In summary, the semi-supervised multi-source domain liveness detection framework proposed in this paper significantly outperforms the original SSDG and IADG based on ResNet-18, achieving performance comparable to DGUA_FAS while maintaining a good balance between accuracy and efficiency. Through continuous innovations in backbone networks, feature fusion, supervision signals, and model compression, SSDG has the potential to further approach or even surpass state-of-the-art liveness detection performance while preserving its lightweight advantages.

4.3 Comparison of Training and Detection Across Different Datasets

To further investigate the model's generalization detection capability under diverse attack scenarios, including traditional attack types and novel AI-based facial attacks, we kept the three source domains (CASIA-FASD, MSU-MFSD, Idiap Replay-Attack) unchanged and only varied the composition of the target domain training data, training two fused models using OULU+AI_dataset as the target domain.

OULU+AI_I: Based on the OULU training set, this dataset incorporates the first batch of AI-generated data (AI_dataset_I), containing 3,203 images (2,202 real faces and 1,001 forged faces).

randomly divided into training set (50%), test set (25%), and validation set (25%).

OULU+AI_II: Based on the OULU training set, incorporating the second batch of AI-generated dataset (AI_dataset_II).

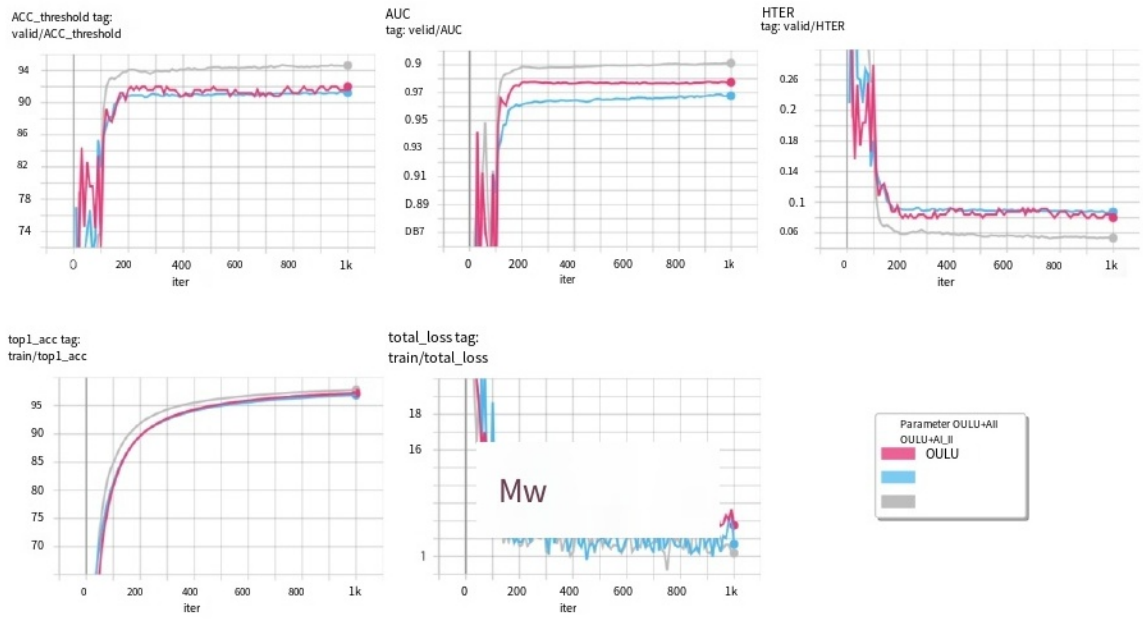


Figure 4-3 The performance of the proposed model after 1,000 iterations on different target domain datasets, evaluated on the training set (second row) and validation set (first row).
Evaluation metric curve plot

All models adopt the same semi-supervised complete configuration (L+E+T+C), with hyperparameters and training strategies consistent with Section 3.3. The final training process is shown in Figure 4-3. It is observed that the proposed model exhibits similar fitting performance across multiple validation iterations during semi-supervised training on target domains with different attack types, demonstrating the model's robustness.

The HTER and AUC of the OULU+AI_I and OULU+AI_II models were evaluated on the AI_dataset_I, AI_dataset_II test sets, and the original OULU test set, with results shown in Table 4-2.

Table 4-2 Performance of the semi-supervised AI facial target domain fusion model on the test set

Semi-supervised training Target domain	HTER of AI_dataset (%)	AUC ofAI_dataset (%)	HTER of OULU (%)	AUC of OULU (%)
OULU+AI_I	6.52	97.58	11.60	95.88
OULU+AI_II	1.30	99.90	14.80	93.25

The following conclusions can be drawn from the training and testing results:

The model trained with semi-supervised learning on the fused target domain demonstrates strong generalization performance against traditional spoofing attacks, adversarial attacks (OULU dataset), and novel AI-generated fake face attacks, consistently achieving a detection accuracy of over 96% on the validation set.

2. The composition of data significantly affects training performance. The OULU+AI_I fused AI_I training set and the OULU training set both contain approximately 1,500 images, meaning that traditional attack types and new AI-based attacks are introduced in similar proportions during training. It was observed that this configuration achieved a low HTER (11.60%) and high AUC (95.88%) on the OULU test set, while maintaining strong performance on the AI dataset as well (HTER = 6.52%, AUC = 97.58%), indicating that this setup effectively balances detection of both conventional presentation attacks and generative forgeries. In contrast, OULU+AI_II performs nearly perfectly on the AI dataset (HTER = 1.13%, AUC = 99.90%), suggesting that the model has learned highly effective discriminative features from this AI dataset. However, its performance drops significantly on the OULU test set, with HTER rising to 14.80% and AUC falling to 93.25%, reflecting weakened generalization capability against physical attacks such as print and replay in OULU. This is because the training set of OULU+AI_II contains images from the AI_II dataset and the OULU dataset in a ratio close to 3:1, causing the model to over-rely on AI-related features during learning.

Highly discriminative features related to data, on the contrary, overlook the more fundamental differences between real faces and physical attacks, leading to overfitting.

Therefore, when using AI-generated data to enhance training, it is important to pay attention to the alignment between the data distribution and the real attack types in the target domain, striving for a balance between diversity and representativeness. Blindly introducing generated samples with significant distributional bias may impair the model's generalization capability on the original task.

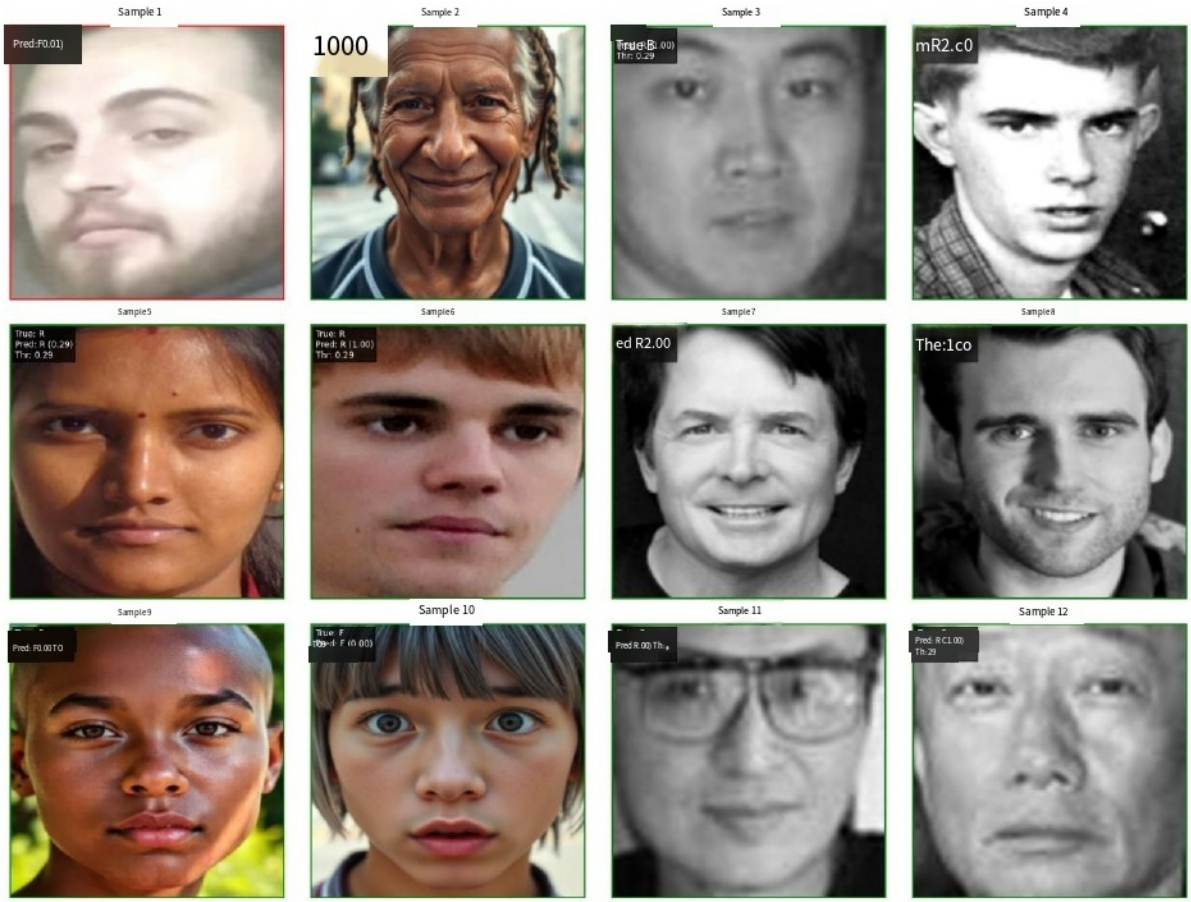


Figure 4-4 shows the prediction results of the proposed model on the test set of the AI_I dataset (green boxes indicate correct predictions, red boxes indicate incorrect predictions), with predicted labels, prediction probabilities, and the optimal model threshold marked.



Figure 4-5 shows the prediction results of the proposed model on the test set of the AI_II dataset (green boxes indicate correct predictions, red boxes indicate incorrect predictions), with predicted labels, prediction probabilities, and the optimal model threshold marked.

Conclusion

This paper addresses the cross-domain generalization problem in facial liveness detection. Building upon an analysis of the strengths and limitations of existing domain generalization and semi-supervised domain adaptation methods, we propose a semi-supervised one-sided adversarial domain generalization framework. Based on SSDG—a one-sided domain generalization method for anti-spoofing proposed in 2020—the framework retains its asymmetric triplet loss and one-sided adversarial mechanism to preserve the inter-domain separation among multi-class spoof samples in the source domain. During data augmentation, we innovatively introduce an image scrambling block consistency loss, enabling the model to focus on local anti-spoofing key features. Furthermore, inspired by the semi-supervised idea of USDAN, we incorporate a small number of labeled samples and a large number of unlabeled samples from the target domain. Through joint optimization using labeled target domain classification loss, entropy minimization loss, domain adversarial loss, and center alignment loss applied exclusively to real samples, the model achieves both strong multi-source domain generalization and precise training with limited target domain data.

This design is the first to effectively utilize unlabeled target domain data in multi-source domain face liveness detection, enabling the model to adaptively reduce distributional differences with the target domain while maintaining strong discriminative power against source domain spoofing samples. Ablation and comparative experiments were conducted using CASIA-FASD, MSU-MFSD, and Idiap Replay-Attack as source domains and OULU-NPU as the target domain. Results show that our method reduces the half total error rate from 26% of the original SSDG to 9. 2%, and increases the AUC to 96. 8%, demonstrating significant enhancement in cross-domain generalization through the semi-supervised component. Further ablation studies on individual components reveal their specific contributions: labeled target domain classification provides direct supervision; entropy minimization encourages high-confidence predictions for unlabeled target samples; domain adversarial loss promotes feature distribution alignment; and center alignment loss, applied only to genuine samples, avoids interference from spoofing samples during domain separation.

Moreover, incorporating AI-generated facial data into the target domain training set still maintains the detection accuracy around 97%, demonstrating the model's strong learning and generalization capabilities. This work not only provides a feasible technical approach for leveraging unlabeled data in the target domain to enhance the generalization performance of face liveness detection, but also offers new insights into integrating domain generalization with semi-supervised domain adaptation. However, compared to state-of-the-art text-prompted multimodal models in the field, our model still has certain gaps. Future research will focus on more advanced backbone networks, more efficient semi-supervised strategies, and unified detection of physical attacks and deepfakes.

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